

Live Demonstration: Real-time Event-based Daily Human Action Recognition with On-chip learning on ANP-I

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Abstract— Real-time human action recognition can be used in applications ranging from elderly care monitoring to context-aware smart homes. However, current action recognition systems face two key challenges: 1) privacy risks proposed by RGB cameras, and 2) low recognition accuracy caused by the complexity and dynamic nature of real-world environments. To address these challenges, we propose a real-time even-based daily human action recognition system with on-chip learning ability. The proposed system utilizes an event camera to capture human action, inherently preserving user privacy. Furthermore, we adopt a lower power neuromorphic processor (ANP-I) and leverages its on-chip learning ability to achieve high-accuracy action recognition. The system is evaluated on six actions—standing, running, jumping, bending, sitting and falling, from daily action dataset, and achieved an overall recognition accuracy of 94%.

Keywords—On-Chip Learning; Event Camera; Human Action Classification

I. DEMONSTRATION SYSTEM

An event camera with 320x320 pixels resolution is used for human action capturing. The event data is sent to a Xilinx ZYNQ platform for preprocessing via SPI interface. After preprocessing, the data is sent to ANP-I neuromorphic processor [1] for real-time inference. A three layers (1024-512-10) spiking neural network (SNN) is implemented on ANP-I. This network performs action recognition based on the spike events from the event camera, eliminating the need to convert spike events into frames. The network performs recognition once every second, and the recognition results are transmitted to a PC through FPGA via the UART interface. Ultimately, all classification results are visualized in real-time through a graphical interface, offering clear and immediate feedback. The detailed data pipeline is illustrated in Figure 1 b).

The proposed system recognizes a set of pre-defined daily actions, such as sit, stand and run. The system features on-chip learning to learn new, custom actions. The on-chip learning energy consumption of ANP-I is small (sub-0.1 μ J/learning), which allows it to be deployed on our system. Moreover, the on-chip learning ability of ANP-I is able to improve the action recognition accuracy from 70% on the DailyAction-DVS dataset [2] to 94%.

II. VISTOR EXPERIENCE

Visitors are invited to interact directly with the system's on-chip learning ability in a hands-on demonstration. The demonstration proceeds through three main stages:

- 1) Custom action collecting: a visitor can define and perform a new, custom action. This stage emphasizes the system's ability to adapt to previously unseen human action.
- 2) On-chip Learning: after data acquisition, the visitor can then witness the learning process in real-time as the neuromorphic chip adapts its parameters to the new action. The on-chip learning ability improves robustness of our system against the dynamic environments.
- 3) Inferencing: to validate the effectiveness of the on-chip learning process, the visitor can immediately perform their new action again and see the recognition results on the GUI, showcasing the on-chip adaptation ability.

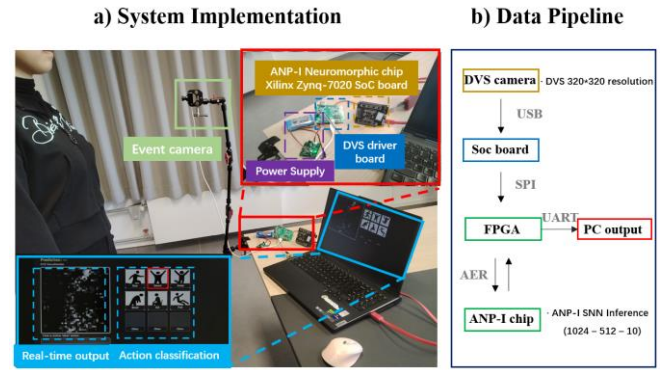


Figure 1: a) System Implementation and b) Data Pipeline.

III. RESULTS

The results of our system are demonstrated live and provide real-time feedback to visitors. When a visitor introduces a new, custom action, the system may initially fail to recognize it, but after a few examples for on-chip training, it will update its parameters and adapts to the new action. This adaptation ability has been validated on the DailyAction-DVS dataset, where it achieves up to 20% accuracy gain in stand, seat and run action classes.

REFERENCES

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